Varun Vinod Singh

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WORK EXPERIENCE

Data & Automation Analyst, Study Space Operations | Northeastern University **Jan 2026 – Present**

Boston, MA · MS Fabric, Delta Lake, PySpark, Dataverse, Fabric Data Pipeline, Power BI, DAX, Python

- Built an hourly Fabric Data Pipeline ingesting **112 daily submissions** from 4 station sources through Dataverse into a Delta Lake, standardizing inconsistent schemas with PySpark.
- Modeled booking, utilization, footfall, and rounds coverage into **7 fact and dimension tables**, loaded through Dataflow Gen2 into a Power BI semantic layer.

Digital Data Engineer Intern | Mass General Brigham **June 2025 – Dec 2025**

Boston, MA · MS Fabric, Fabric Data Pipelines, PySpark, MySQL, ServiceNow, Power BI, Python, SQL

- Consolidated ServiceNow data from MySQL into an analytics-ready Fabric model through a scheduled data pipeline, joining and cleaning multiple source tables for **3 clinical operations teams**.
- Anonymized patient PHI across **100,000+ records** with PySpark, joining demographic and clinical tables to feed an AI model extracting key findings from doctor prescriptions.

Software Engineer | LTM Limited (formerly LTIMindtree) **Sep 2022 – Jul 2024**

Mumbai, India · Python, Java, Kafka, Spark, Airflow, AWS S3, Glue Data Catalog, Parquet, SQL

- Engineered 3 streaming Kafka and Spark pipelines in Python and Java, landing 5 ServiceNow sources as partitioned Parquet on S3 at **1,500 records per minute**, orchestrated as Airflow DAGs with retries, SLA alerts, and backfills.
- Designed a modular validation layer of schema rules, type checks, and deduplication over Glue Data Catalog tables, failing the DAG before publish and cutting data incidents **45%, from 11 to 6 a month**.

PROJECTS

NYC Transit Data Platform | *Kafka, Airflow, Snowflake, dbt, Databricks, Streamlit*

- Architected a streaming lakehouse ingesting **100M+ NYC transit records** through Kafka into Databricks Delta tables, modeled with dbt into Snowflake marts serving a Streamlit dashboard.
- **I**mplemented modular dbt models over a Bronze/Silver/Gold medallion design, with incremental loads trimming each run from **25 to 15 minutes, 40% faster.**

VectorForge | *Apache Airflow, Docker, Postgres, pgvector, pandera, MinIO, sentence-transformers*

- Orchestrated a **4-DAG Airflow platform** in Docker, landing arXiv data as immutable partitioned JSON in S3-compatible storage and gating Postgres loads on pandera schema validation.
- Embedded chunked abstracts into pgvector with an IVFFlat cosine index, enforcing a **3-day freshness SLA** and logging run counts, staleness, and status to a queryable health table.

LLM-Powered SEC Filing Search Engine | *Python, LangChain, prompt engineering, embeddings, fastembed, Streamlit*

- Deployed an LLM-powered RAG application applying NLP to chunk, embed, and retrieve across 500+ SEC filings (PDF, HTML), serving cited natural-language answers end-to-end through a live user-facing interface.
- Developed a RAGAS-based validation framework to monitor agentic pipeline outputs on faithfulness and answer relevance, tuning chunk size and top-k to improve retrieval quality.

TECHNICAL SKILLS

Model & Transformation: dbt, dimensional modeling, medallion architecture, semantic layer, data marts, partitioning
Warehouses & Databases: Snowflake, Databricks, Delta Lake, Postgres, MySQL, pgvector
Orchestration & Pipelines: Apache Airflow, Fabric Data Pipelines, Kafka, PySpark, Apache Spark, ETL/ELT, batch, streaming
Cloud & Infrastructure: AWS, S3, Glue Data Catalog, Azure, MS Fabric, Azure Data Factory, Docker, Parquet
Languages: Python, SQL, Java, Bash, Git
Data Quality & Reliability: pandera, schema validation, deduplication, freshness SLA, idempotency, run observability
AI, Retrieval & BI: RAG, embeddings, vector search, NLP, LangChain, RAGAS, Power BI, DAX, Streamlit

EDUCATION

Northeastern University, M.S. Information Systems **December 2026**
Mumbai University, B.E. Information Technology **May 2022**